



Featured Letter

Preparation of a self-healing silicone coating for inhibiting adhesion of benthic diatoms

Miaosen Yang^{a,b}, Yuhang Sun^{a,b}, Guangmeng Chen^{a,c}, Guangyao Wang^{a,b}, Songzhu Lin^{a,*}, Zhiyong Sun^{b,*}^a School of Chemical Engineering, Northeast Electric Power University, Jilin 132012, PR China^b State Key Laboratory for Marine Corrosion and Protection, Luoyang Ship Material Research Institute (LSMRI), Qingdao 266273, PR China^c School of Mechanics and Civil & Architecture, Northwestern Polytechnical University, Xi'an, PR China

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ABSTRACT

This work reports a silicone coating (PDMS-Ula) with marine antifouling and self-healing properties, which was prepared by the first synthesis of polydimethylsiloxane polymer containing imine and urea groups and then adding a small amount of eco-friendly antifoulant. The characteristic structure of the polymer was verified via Fourier transform infrared spectroscopy (FT-IR) and nuclear magnetic resonance (NMR). The PDMS-Ula coating has rapid self-repairing performance, excellent transparency and the ability to control the release of antifouling agents at a certain rate. The adhesion experiment using four typical benthic diatoms indicated that the PDMS-Ula coating can effectively inhibit the attachment of diatoms, which effectively improves the static antifouling ability of silicone materials.

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1. Introduction

Biofouling or fouling, the settlement and growth of marine organisms containing bacteria, algae and crustacean on any structure surfaces immersed in seawater, has always raised severe problems. It accelerates biocorrosion, increases roughness and therefore results in extra fuel consumption, causes species invasion and harms the marine aquaculture industry [1]. Currently, based on the requirements of environmental protection and long-acting antifouling, using the natural characteristics of materials to inhibit the adhesion of marine organisms has attracted considerable research interest and is very promising. For example, poly (dimethylsiloxane) (PDMS), an organosilicon elastomer which possesses low surface energy and elastic modulus, makes marine organisms only weakly adhere to the surface and be readily removed by shear forces [2]. Unfortunately, the PDMS elastomers are only effective when the ship is sailing at a high speed (15 > knots), and they exhibit poor antifouling performance on static conditions because of their high interfacial energy with water that favors the adsorption of biomolecules, diatoms and sludge [3]. This limits the application of silicone coatings under static conditions, where there are no hydrodynamic shear forces to remove settled organisms. Moreover, PDMS elastomers easily suffer dam-

age in the process of usage owing to their poor mechanical properties, which not only causes shortened life but also results in safe problems. Even worse, it is difficult to repair once damaged due to their cross-linked network structure. Besides, the transparency of antifouling materials is a very important property which makes the antifouling materials apply to the protection of specialized devices in seawater, such as subsea windows, light sources and optical instrument. Thus, to solve the above problems, designing and developing novel PMDS-based materials with excellent static antifouling, transparency and self-healing capability is extremely meaningful.

Today, the self-repairing technology is usually driven by dynamic covalent bonds (imine bond [4], disulfide bond [5] and acylhydrazone bond [6], etc.), supramolecular interactions (hydrogen bonding [7] and π - π stacking [8]) or incorporating microcapsules into polymers [9]. Among them, imine bands formed by the condensation of amino and carbonyl functional groups, present outstanding dynamic characteristics that can be activated under mild conditions [10]. Therefore, this inspired us to develop novel self-healing PDMS-based antifouling materials with the help of dynamic imine bonds.

In this work, a silicone coating composed of imine-containing PDMS-based polymer and a small amount of N-octyl-2-hydroxybenzamide was developed by taking advantage of Schiff-base reaction and an eco-friendly antifouling agent. The structural features and the properties including self-healing, transparency and antifouling for the coating were discussed.

* Corresponding authors.

E-mail addresses: linsongzhu@neepu.edu.cn (S. Lin), szy_sunrui725@163.com (Z. Sun).

2. Experimental section

2.1. Materials

Aminopropyl terminated polydimethylsiloxane was purchased (PDMS, MW = 2000 g/mol) from Hangzhou Silong Mater. Tech. Co., Ltd. Isophthalaldehyde (IPAL) and 4,4'-methylene bis-(cyclohexyl isocyanate) (HMDI) were purchased from Macklin and sigma, respectively. N-octyl-2-hydroxybenzamide and tetrahydrofuran (THF) provided by Sinopharm Chemical Reagent Co., Ltd.

2.2. Synthesis of the self-healing coatings (PDMS-UIa)

PDMS (0.01 mol) and THF (100 mL) were added to a 250 mL three-necked round-bottom flask equipped with a reflux device. When the system was heated to 90 °C in an oil bath under an N₂ atmosphere, 0.0055 mol HMDI was dropwise poured to the mixture with vigorous stirring for 24 h. Subsequently, 0.0045 mol IPAL was added and stirred for 3 h at 60 °C. Afterwards, adding 1.10 g antifoulant to the system and stirring for 30 min. Finally, the

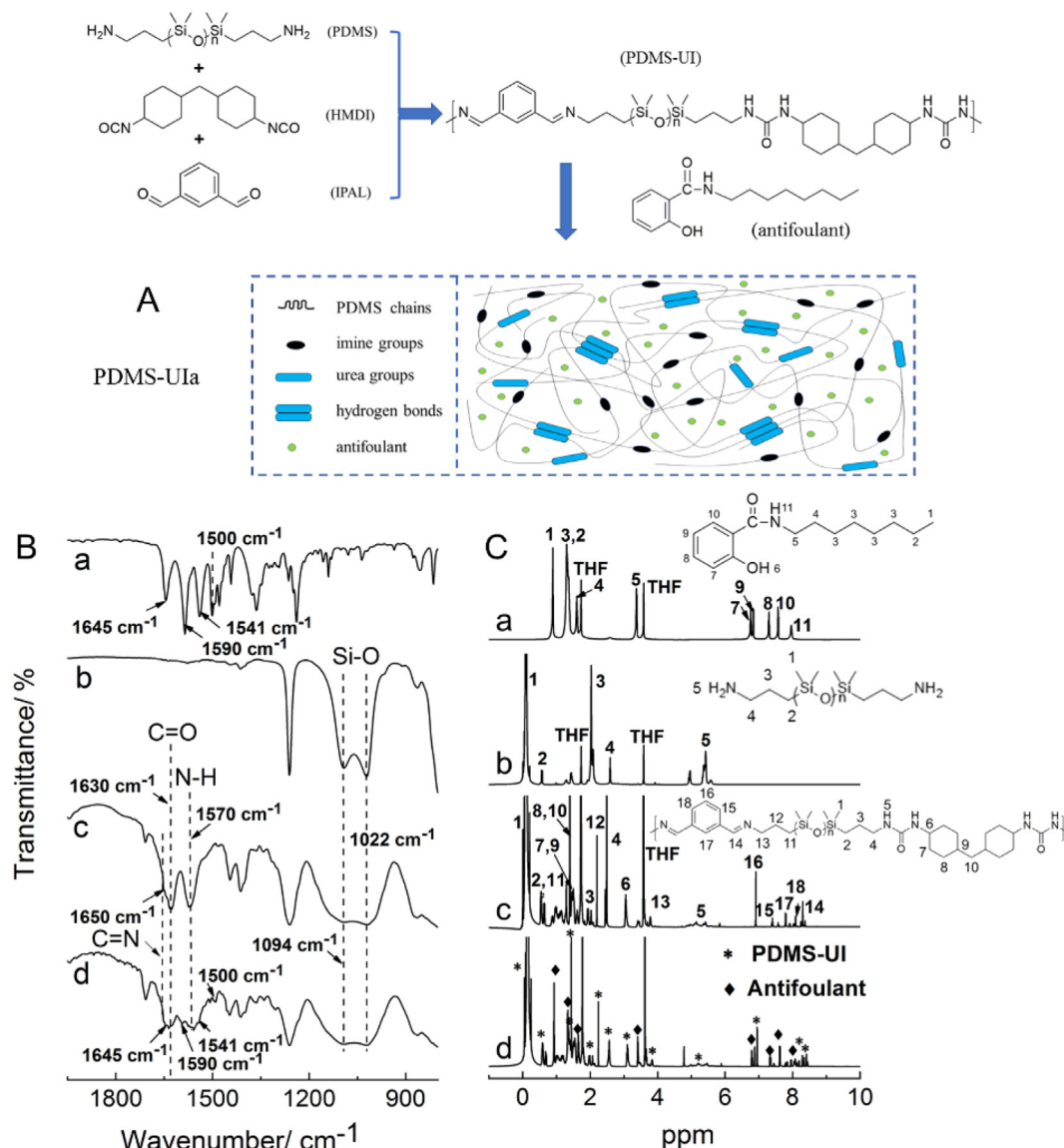
self-healing coatings (PDMS-UIa) were obtained by applying it on slide glass with an area of 2.5 cm × 2.5 cm.

2.3. The experimental methods of coating self-healing

The coatings were first immersed into 2.0%, 3.0%, 3.5% and 4.0% NaCl aqueous solution respectively and then scratched a crack artificially using a scalpel upon the coating surface. The self-repairing of the coating surface was recorded by the Hirox digital microscope KH-8700 every an hour.

2.4. The release test and determination of diatoms adhesive

The release test of antifoulant was performed at 20 °C by immersing the coatings in 200 mL of 3.5% NaCl solution stirred with a speed of 500 r/min. The released antifoulant was measured every 24 h by UV-vis absorption spectrum. Four diatoms including *Navicula leavisima*, *Navicula pinna*, *Chaetoceros socialis* and *Phaeodactylum tricornutum* Bohlin were employed to evaluate the anti-adhesive performance of the coating. A 10 mL diatom suspension



with the concentration of 1.5×10^6 cells/ml was settled on the coatings for 2 h at 20 °C. Unsettled diatoms were gently washed in seawater. Diatom amount on the coating after exposure to flow was recorded using the Hirox digital microscope KH-8700. Diatoms were counted in 10 fields of view (0.059 mm^2 each) in each of three parallel coatings to provide cell settlement data.

2.5. Characterization

FT-IR spectra were recorded on a Nicolet 8700 Fourier transfer infrared spectrophotometer using the KBr disk method. ^1H Nuclear magnetic resonance ($^1\text{H-NMR}$) spectroscopy was recorded on a Bruker AVANCE III (600 MHz) in THF at room temperature. The scanning Kelvin probe (SKP) measurements were performed by an AMETEK VersaScan electrochemistry workstation stepped over a designated area ($2000 \mu\text{m} \times 2000 \mu\text{m}$) of a sample surface. The transparency of the coating painted on a quartz plate with an area

of $1 \text{ cm} \times 3 \text{ cm}$ was determined by a UV-visible absorption spectrometer (Hitachi U-2800) in the range of 300–800 nm with the slit width of 1.0 nm.

3. Results and discussion

The brief synthetic processes of PDMS-Ula are shown in Fig. 1A. Firstly, PDMS-UI was prepared by the sequential reaction of amino-propyl polydimethylsiloxane (PDMS) with 4,4'-methylenebis(cyclo hexyl isocyanate) (HMDI) and isophthalaldehyde (IPAL). Then a certain amount of antifouling agent was added to PDMS-UI, stirring and curing at room temperature to obtain PDMS-Ula. The structural information of PDMS-Ula, as well as antifoulant, PDMS, and PDMS-UI, was revealed by FT-IR spectroscopy and ^1H NMR spectrum (Fig. 1B and C). The spectra of antifoulant (Fig. 1B-a) shows two strong absorption bands at 1645 cm^{-1} and 1541 cm^{-1} that can be attributed to the stretching vibration of C=O and C-N,

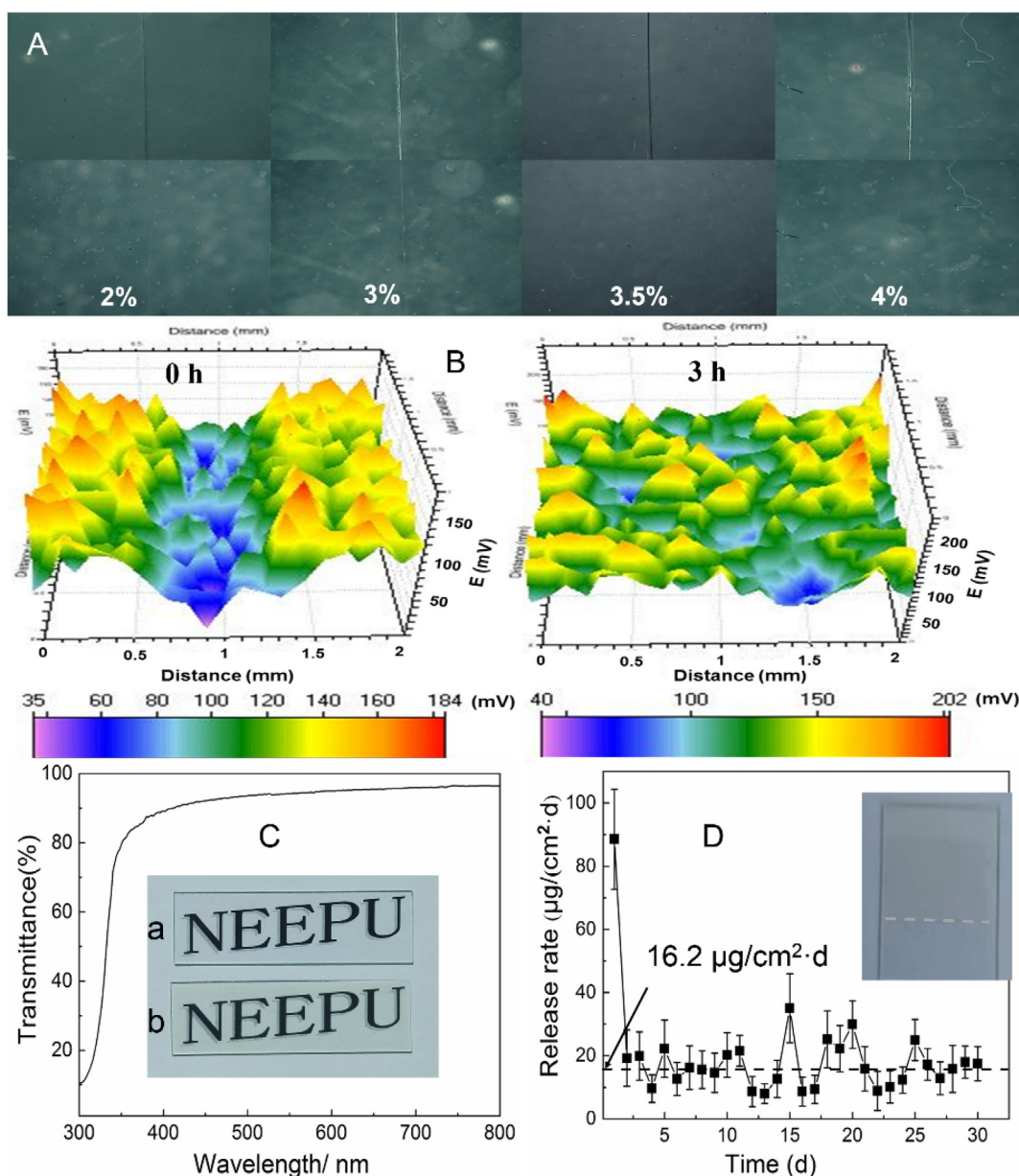


Fig. 2. (A) Optical photos of the PDMS-Ula crack before and after healing 3 h in 2.0%, 3.0%, 3.5% and 4.0% NaCl solution. (B) SKP images of the damaged PDMS-Ula coating before and after immersion in 3.5% NaCl solution for 3 h. (C) The optical property of PDMS-Ula coated on a quartz plate (inserts: a, naked glass slide; b, PDMS-Ula coated on a glass slide). (D) The release profile of antifoulant from the PDMS-Ula coating in NaCl solution (3.5%). Indicated values are means of three measurements.

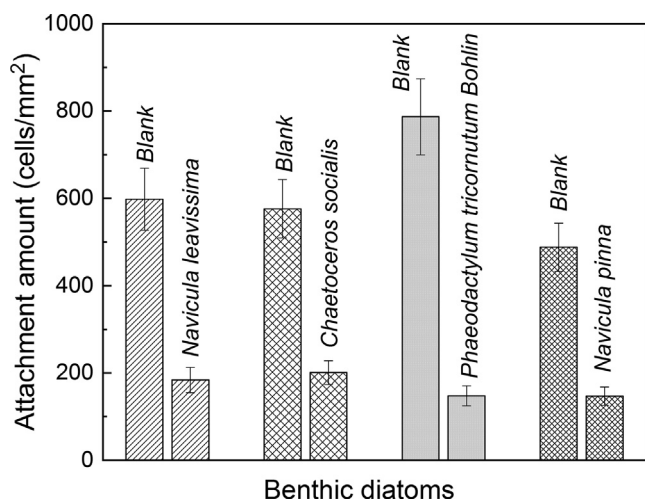


Fig. 3. The numbers of four benthic diatoms attached to the surface of PDMS-UI (blank) and PDMS-UIa coatings respectively.

respectively. The characteristic absorption peaks at 1590 cm^{-1} and 1500 cm^{-1} are attributed to the benzenes vibration. In the spectrum of PDMS (Fig. 1B-b), the strong and broad absorption bands in the region $1022\text{--}1094\text{ cm}^{-1}$ are due to the Si-O stretching vibration of the silicon frameworks. For the PDMS-UI (Fig. 1B-c), the appearance of the C=O stretching vibration at 1630 cm^{-1} and N-H bending vibration of secondary amine at 1570 cm^{-1} indicates the existence of urea groups. The C=N stretching vibration peak also was observed at 1650 cm^{-1} , illustrating the formation of imine bonds. As expected, the characteristic bands corresponding to PDMS-UI and antifoulant molecules were observed in PDMS-UIa sample (Fig B-d).

A complete assignment of ^1H NMR resonances for antifoulant and PDMS was performed, and the results are shown in Fig. 1C-a and -b (antifoulant: H1 0.90, H2, H3 1.31, H4 1.61, H5 3.37, H7 6.71, H8 7.29, H9 6.86, H10 7.55, H11 7.94 ppm; PDMS: H1 0.11, H2 0.58, H3 1.99, H4 2.57, H5 4.97, 5.44 ppm). In the case of PDMS-UI (Fig. 1C-c), the characteristic signal peaks of urea and imine groups were observed such as secondary amine group (4 ~ 6 ppm), methylene and methine next to the urea groups (2.53 and 3.05 ppm), benzene ring groups (6.8–8.1 ppm), and imine groups (8.31 ppm). Combined with FT-IR results, it has been confirmed that the self-healing silicone material with the designed structure was successfully synthesized. Likewise, PDMS-UIa possesses the characteristic structural information containing PDMS-UI and antifoulant agent (Fig. 1C,d).

To demonstrate the self-healing properties of PDMS-UIa, the scratch test was carried out in 2.0%, 3.0%, 3.5% and 4.0% NaCl solution at room temperature, respectively. As shown in Fig. 2A, the crack almost disappeared after 3 h. The healing details of PDMS-UIa in 3.5% NaCl solution were also revealed by Scanning Kelvin probe technique (Fig. 2B), which the damaged surface is completely restored, indicating its excellent self-healing ability in seawater. Besides, PDMS-UIa exhibits excellent light transmission in the visible light region, which the light transmittance is over 90% (Fig. 2C). In addition, since the release rate of antifoulant plays a critical role in marine anti-biofouling, the release experiment of antifoulant is performed for 30 days, as shown in Fig. 2D. It was found that the release rate of antifoulant has a certain fluctuation,

and the average release rate is $16\text{ }\mu\text{g/cm}^2\text{ d}$. This means that silicone self-healing materials can control the release of antifouling agents at a slightly fluctuating rate, which benefits the long-term antifouling under the static conditions.

The adhesion experiment using *Navicula leavisissima*, *Navicula pinna*, *Chaetoceros socialis* and *Phaeodactylum tricornutum Bohlin* on the surface of the PDMS-UIa coating was performed to evaluate its antifouling capacities, as well as the PDMS-UI coating for comparison. Fig. 3 shows the numbers of four diatoms attached to the PDMS-UI and PDMS-UIa coatings. As expected, the PDMS-UIa coating effectively inhibits the settlement of four typical benthic diatoms in comparison with the PDMS-UI coating, demonstrating its excellent static antifouling properties.

4. Conclusion

In conclusion, We have developed a self-healing coating for marine antifouling that is inspired by a combination of two antifouling mechanisms, namely the antifouling release of silicone materials and the killing or repellency of antifouling agents. The coating exhibits the fast self-repairing property, superior transparency and ability to control the release of antifoulant. Moreover, the coating can significantly inhibit the adhesion of four typical benthic diatoms, indicating a good static antifouling ability, and is promising in the field of marine antifouling.

CRedit authorship contribution statement

Miaosen Yang: Conceptualization, Data curation, Writing - original draft, Funding acquisition. **Yuhang Sun:** Investigation. **Guangmeng Chen:** Validation. **Guangyao Wang:** Resources. **Songzhu Lin:** Validation, Project administration. **Zhiyong Sun:** Methodology, Writing - review & editing, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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